JORGE SANTANA MENDOZA

EMBEDDED SOFTWARE ENGINEER

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SUMMARY

Embedded Software Engineer with 6+ years of experience in automotive embedded systems, specializing in software verification, source-level debugging, HIL testing, and test automation. Experienced in diagnosing ECU software issues, developing and validating patches, and collaborating with international teams and software suppliers. Strong background in infotainment and ADAS, including verification strategy and test coordination.

EDUCATION

- **Master's in Applied Artificial Intelligence**
  Tecnológico de Monterrey | Mexico
  2024 - 2026
- **Bachelor's Degree in Mechatronics Engineering**
  Tecnológico de Monterrey | Mexico
  2015 - 2020 (Honors - GPA: 92.5)
- **International Program**
  Hochschule Karlsruhe | Germany
  2018 - 2019

TECHNICAL SKILLS

- **Testing & Automation Tools:**
  - CANoe, vTESTstudio, Robot Framework
- **Embedded Operating Systems & Platforms:**
  - Linux, QNX
- **Programming & Scripting Languages:**
  - Python, CAPL
- **Development & CI/CD Tools:**
  - Docker, Jenkins
- **Version Control & Code Review:**
  - Git, GitHub, Gerrit, Bitbucket
- **Communication Protocols:**
  - CAN, CAN-TP, CAN FD, XCP, Ethernet, UDS
- **Model-Based Design & Simulation:**
  - Simulink, Stateflow
- **Project & Requirements Management:**
  - Jira, DOORS/RQM, Confluence
- **Automotive Standards & Processes:**
  - SDLC, ISTQB, ASPICE, AUTOSAR

SOFT SKILLS

- Problem Solving
- Leadership
- Proactivity
- Cross-functional Collaboration

LANGUAGES

- English - Upper-intermediate (B2)
- German - Intermediate (B1)

CERTIFICATIONS

- **IBM AI Engineering Professional Certificate**
  IBM(2025)
- **Scrum Master Professional Certification**
  Certiprof (2025)
- **ISTQB® Certified Tester Foundation Level**
  Certification Brightest: 2024028872 (2024)

PROFESSIONAL EXPERIENCE

Robert Bosch

Embedded Software Engineer | 07/2025 - Present

Software verification and debugging of infotainment ECU software on HIL platforms.

- Develop and execute automated verification tests using Robot Framework.
- Perform source-level debugging, including software patch development, build generation, and targeted validation.
- Collaborate directly with QNX and Qualcomm engineering teams to investigate platform software defects, share technical findings, and validate proposed fixes.

Embedded Software Test Function Lead | 02/2025 - 07/2025

Leadership and coordination of an international engineering team responsible for system testing of an ADAS module across the United States, India, and Mexico.

- Defined the overall verification strategy for the software function block.
- Specified, executed, and analyzed system tests using CANoe and vTESTstudio.
- Defined and tracked test activities against project milestones.
- Reported test status and findings to stakeholders.
- Onboarded and mentored new team members across different locations.

Embedded Software Test Engineer | 09/2022 - 02/2025

System verification of an ADAS ECU on HIL platforms.

- Performed system testing of software features: flashing, communication, diagnostics, fault management, state management, robustness, and runtime performance.
- Automated test scenarios using vTestStudio and CANoe.
- Analyzed system requirements, defined test specifications, and implemented automated test scenarios using Python and CAPL.
- Created and tracked defect reports with Jira and RQM.

ALTEN Mexico

Software Test Engineer Jr. at Volkswagen | 07/2021 - 09/2022

System validation of infotainment and ADAS ECUs on vehicle and HIL platforms.

- Built and maintained HIL test benches for different vehicle architectures.
- Validated infotainment features, including media playback, navigation, Android Auto and Apple CarPlay integration, phone connectivity, and radio.
- Tested ADAS functions, including ACC, lane assist, and park assistance.

HIL Test Engineer Jr. at Ford Motor Company | 08/2020 - 07/2021

Development of Simulink/Stateflow models of ECUs for hardware-in-the-loop testing.

- Developed ECU models in Simulink based on technical specifications and CAN data.
- Integrated and validated ECU models with cross-functional software and hardware teams.